Question 12.118

AI25BTECH11040 - Vivaan Parashar

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1 Question:

One of the eigenvalues of a 3×3 matrix $\mathbf{M}$ is 3. If the determinant of the matrix $\mathbf{M}$ is 24 and the trace is 9, then the smallest eigenvalue of the matrix $\mathbf{M}^{-1}$ is

a) 1/8

b) 1/4

c) 1/3

d) 1/2

2 Solution:

Let the eigenvalues of the matrix $\mathbf{M}$ be λ_1, λ_2, and λ_3. Given that one of the eigenvalues is 3, we can denote:

$$\lambda_1 = 3$$

The determinant of the matrix $\mathbf{M}$ is given by the product of its eigenvalues:

$$\det(\mathbf{M}) = \lambda_1 \cdot \lambda_2 \cdot \lambda_3 = 24$$

This is because the characteristic polynomial (for a 3×3 matrix) is:

$$P(\lambda) = \det|\mathbf{M} - \lambda \mathbf{I}_3| = -(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3)$$

Clearly setting $\lambda = 0$ gives the determinant as the product of eigenvalues. Substituting the known eigenvalue:

$$3 \cdot \lambda_2 \cdot \lambda_3 = 24$$

$$\lambda_2 \cdot \lambda_3 = \frac{24}{3} = 8 \tag{1}$$

The trace of the matrix $\mathbf{M}$ is given by the sum of its eigenvalues (as can be derived from the characteristic polynomial):

$$\mathrm{tr}(\mathbf{M}) = \lambda_1 + \lambda_2 + \lambda_3 = 9$$

Substituting the known eigenvalue:

$$3 + \lambda_2 + \lambda_3 = 9$$

$$\lambda_2 + \lambda_3 = 6$$

Now we have a system of equations:

$$\lambda_2 + \lambda_3 = 6 \tag{2}$$
$$\lambda_2 \cdot \lambda_3 = 8 \tag{3}$$

We can solve this system by treating λ_2 and λ_3 as the roots of the quadratic equation:

$$x^2 - (\lambda_2 + \lambda_3)x + \lambda_2 \cdot \lambda_3 = 0$$

Substituting the values from equations (3) and (4):

$$x^2 - 6x + 8 = 0$$

This gives us two solutions:

$$x_1 = 4 \tag{4}$$
$$x_2 = 2 \tag{5}$$

Thus, the eigenvalues of the matrix $\mathbf{M}$ are:

$$\lambda_1 = 3, \quad \lambda_2 = 4, \quad \lambda_3 = 2$$

The eigenvalues of the inverse matrix $\mathbf{M}^{-1}$ are the reciprocals of the eigenvalues of $\mathbf{M}$:

$$\mu_1 = \frac{1}{\lambda_1} = \frac{1}{3}, \quad \mu_2 = \frac{1}{\lambda_2} = \frac{1}{4}, \quad \mu_3 = \frac{1}{\lambda_3} = \frac{1}{2}$$

The smallest eigenvalue of the matrix $\mathbf{M}^{-1}$ is:

$$\min\left(\frac{1}{3}, \frac{1}{4}, \frac{1}{2}\right) = \frac{1}{4}$$

Thus, the smallest eigenvalue of the matrix $\mathbf{M}^{-1}$ is:

$$\boxed{\frac{1}{4}}$$